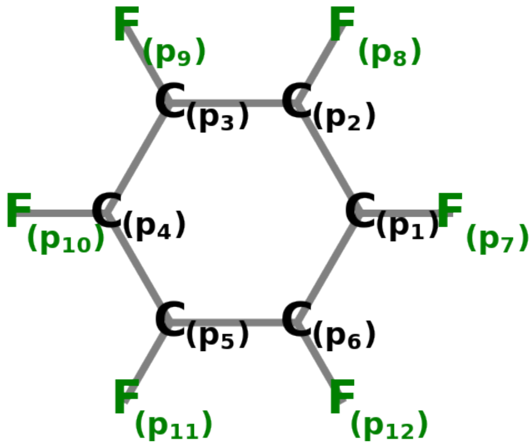
Hexafluorobenzene



Point Group: **D6h**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 12E \oplus -4C'2 \oplus -12\sigma h \oplus 4\sigma v$$

$$\Gamma_{irred} = 2B2g \oplus 2E1g \oplus 2A2u \oplus 2E2u$$

- SALC of B2g: $\frac{\sqrt{6}(\mathrm{p}_1 - \mathrm{p}_2 + \mathrm{p}_3 - \mathrm{p}_4 + \mathrm{p}_5 - \mathrm{p}_6)}{6}$
- SALC of B2g: $\frac{\sqrt{6}(-\mathrm{p}_{10} + \mathrm{p}_{11} - \mathrm{p}_{12} + \mathrm{p}_7 - \mathrm{p}_8 + \mathrm{p}_9)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(2\mathrm{p}_1 + \mathrm{p}_2 - \mathrm{p}_3 - 2\mathrm{p}_4 - \mathrm{p}_5 + \mathrm{p}_6)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(\mathrm{p}_1 + 2\mathrm{p}_2 + \mathrm{p}_3 - \mathrm{p}_4 - 2\mathrm{p}_5 - \mathrm{p}_6)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(-\mathrm{p}_1 + \mathrm{p}_2 + 2\mathrm{p}_3 + \mathrm{p}_4 - \mathrm{p}_5 - 2\mathrm{p}_6)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(-\mathrm{p}_{10} - 2\mathrm{p}_{11} - \mathrm{p}_{12} + \mathrm{p}_7 + 2\mathrm{p}_8 + \mathrm{p}_9)}{6}$
- SALC of A2u: $\frac{\sqrt{6}(\mathrm{p}_1 + \mathrm{p}_2 + \mathrm{p}_3 + \mathrm{p}_4 + \mathrm{p}_5 + \mathrm{p}_6)}{6}$
- SALC of A2u: $\frac{\sqrt{6}(\mathrm{p}_{10} + \mathrm{p}_{11} + \mathrm{p}_{12} + \mathrm{p}_7 + \mathrm{p}_8 + \mathrm{p}_9)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(2\mathrm{p}_1 - \mathrm{p}_2 - \mathrm{p}_3 + 2\mathrm{p}_4 - \mathrm{p}_5 - \mathrm{p}_6)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(-\mathrm{p}_1 + 2\mathrm{p}_2 - \mathrm{p}_3 - \mathrm{p}_4 + 2\mathrm{p}_5 - \mathrm{p}_6)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(-\mathrm{p}_1 - \mathrm{p}_2 + 2\mathrm{p}_3 - \mathrm{p}_4 - \mathrm{p}_5 + 2\mathrm{p}_6)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(\mathrm{p}_{10} + \mathrm{p}_{11} - 2\mathrm{p}_{12} + \mathrm{p}_7 + \mathrm{p}_8 - 2\mathrm{p}_9)}{6}$

Put together these SALCs form the following Hamilton matrices:

$$H_{B2g} = \left[\begin{bmatrix} \tilde{\alpha} - 2\tilde{\beta} & 0 \\ 0 & \tilde{\alpha}_F \end{bmatrix}\right]$$

$$H_{E1g} = \left[\begin{bmatrix} \tilde{\alpha} + \tilde{\beta} & \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & -\frac{\tilde{\alpha}}{2} - \frac{\tilde{\beta}}{2} & 0 \\ \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} + \tilde{\beta} & \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & 0 \\ -\frac{\tilde{\alpha}}{2} - \frac{\tilde{\beta}}{2} & \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} + \tilde{\beta} & 0 \\ 0 & 0 & 0 & \tilde{\alpha}_F \end{bmatrix}\right]$$

$$H_{A2u} = \left[\begin{bmatrix} \tilde{\alpha} + 2\tilde{\beta} & 6\tilde{\beta}_F \\ 6\tilde{\beta}_F & \tilde{\alpha}_F \end{bmatrix}\right]$$

$$H_{E2u} = \left[\begin{bmatrix} \tilde{\alpha} - \tilde{\beta} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & 0 \\ -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} - \tilde{\beta} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & 0 \\ -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} - \tilde{\beta} & 0 \\ 0 & 0 & 0 & \tilde{\alpha}_F \end{bmatrix}\right]$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A2u with energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_F}{2} + \tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}{2}$
  (
  $$\left[-\frac{12\tilde{\beta}_F}{\sqrt{144\left|\frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}\right|^2 + 1}\left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}\right)} \quad \sqrt{144\left|\right.}$$
  $$= \frac{\sqrt{6}\left(-12\tilde{\beta}_F\left(\mathrm{p}_1 + \mathrm{p}_2 + \mathrm{p}_3 + \mathrm{p}_4 + \mathrm{p}_5 + \mathrm{p}_6\right) + \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} +}\right.\right.}{6\sqrt{144\left|\frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}\right|^2 + 1}\left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F}\right.}$$
)
- orbital of A2u with energy = $\frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_F}{2} + \tilde{\beta} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}{2}$
  (
  $$\left[-\frac{12\tilde{\beta}_F}{\sqrt{144\left|\frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}\right|^2 + 1}\left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}\right)} \quad \sqrt{144\left|\right.}$$
  $$= \frac{\sqrt{6}\left(-12\tilde{\beta}_F\left(\mathrm{p}_1 + \mathrm{p}_2 + \mathrm{p}_3 + \mathrm{p}_4 + \mathrm{p}_5 + \mathrm{p}_6\right) + \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} +}\right.\right.}{6\sqrt{144\left|\frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}\right|^2 + 1}\left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F}\right.}$$
)
- orbital of E1g with energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
  (
  $$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in E1g} \\ \text{2. SALC in E1g} \\ \text{3. SALC in E1g} \\ \text{4. SALC in E1g} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{3}\mathrm{p}_1}{3} + \frac{\sqrt{3}\mathrm{p}_2}{6} - \frac{\sqrt{3}\mathrm{p}_3}{6} - \frac{\sqrt{3}\mathrm{p}_4}{3} - \frac{\sqrt{3}\mathrm{p}_5}{6} + \\ \frac{\sqrt{3}\mathrm{p}_1}{6} + \frac{\sqrt{3}\mathrm{p}_2}{3} + \frac{\sqrt{3}\mathrm{p}_3}{6} - \frac{\sqrt{3}\mathrm{p}_4}{6} - \frac{\sqrt{3}\mathrm{p}_5}{3} - \\ -\frac{\sqrt{3}\mathrm{p}_1}{6} + \frac{\sqrt{3}\mathrm{p}_2}{6} + \frac{\sqrt{3}\mathrm{p}_3}{3} + \frac{\sqrt{3}\mathrm{p}_4}{6} - \frac{\sqrt{3}\mathrm{p}_5}{6} - \\ -\frac{\sqrt{3}\mathrm{p}_{10}}{6} - \frac{\sqrt{3}\mathrm{p}_{11}}{3} - \frac{\sqrt{3}\mathrm{p}_{12}}{6} + \frac{\sqrt{3}\mathrm{p}_7}{6} + \frac{\sqrt{3}\mathrm{p}_8}{3} + \end{bmatrix}$$
  $$= \frac{\sqrt{6}\left(\mathrm{p}_1 + \mathrm{p}_2 - \mathrm{p}_4 - \mathrm{p}_5\right)}{4}$$
)
  Error in Orthogonalization of SALCs
- orbital of E1g with energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
  (
  $$\begin{bmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in E1g} \\ \text{2. SALC in E1g} \\ \text{3. SALC in E1g} \\ \text{4. SALC in E1g} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{3}\mathrm{p}_1}{3} + \frac{\sqrt{3}\mathrm{p}_2}{6} - \frac{\sqrt{3}\mathrm{p}_3}{6} - \frac{\sqrt{3}\mathrm{p}_4}{3} - \frac{\sqrt{3}\mathrm{p}_5}{6} + \\ \frac{\sqrt{3}\mathrm{p}_1}{6} + \frac{\sqrt{3}\mathrm{p}_2}{3} + \frac{\sqrt{3}\mathrm{p}_3}{6} - \frac{\sqrt{3}\mathrm{p}_4}{6} - \frac{\sqrt{3}\mathrm{p}_5}{3} - \\ -\frac{\sqrt{3}\mathrm{p}_1}{6} + \frac{\sqrt{3}\mathrm{p}_2}{6} + \frac{\sqrt{3}\mathrm{p}_3}{3} + \frac{\sqrt{3}\mathrm{p}_4}{6} - \frac{\sqrt{3}\mathrm{p}_5}{6} - \\ -\frac{\sqrt{3}\mathrm{p}_{10}}{6} - \frac{\sqrt{3}\mathrm{p}_{11}}{3} - \frac{\sqrt{3}\mathrm{p}_{12}}{6} + \frac{\sqrt{3}\mathrm{p}_7}{6} + \frac{\sqrt{3}\mathrm{p}_8}{3} \end{bmatrix}$$
  $$= \frac{\sqrt{6}\left(-\mathrm{p}_1 + \mathrm{p}_3 + \mathrm{p}_4 - \mathrm{p}_6\right)}{4}$$
)
  Error in Orthogonalization of SALCs

- orbital of E1g with energy = 0
  (

$$\begin{bmatrix}\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E1g}\\ \text{2. SALC in E1g}\\ \text{3. SALC in E1g}\\ \text{4. SALC in E1g}\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \cdots\\ \frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} - \cdots\\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} + \frac{\sqrt{3}p_4}{6} - \cdots\\ -\frac{\sqrt{3}p_{10}}{6} - \frac{\sqrt{3}p_{11}}{3} - \frac{\sqrt{3}p_{12}}{6} + \frac{\sqrt{3}p_7}{6} + \cdots\end{bmatrix}$$

$$= 0$$

)

  Error in Orthogonalization of SALCsError in Orthogonalization of SALCsError in Orthogonalization of SALCs

- orbital of E1g with energy = $\tilde{\alpha}_F$
  (

$$\begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E1g}\\ \text{2. SALC in E1g}\\ \text{3. SALC in E1g}\\ \text{4. SALC in E1g}\end{bmatrix} = \begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} + \frac{\sqrt{3}p_6}{6}\\ \frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{3} - \frac{\sqrt{3}p_6}{6}\\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} + \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} - \frac{\sqrt{3}p_6}{3}\\ -\frac{\sqrt{3}p_{10}}{6} - \frac{\sqrt{3}p_{11}}{3} - \frac{\sqrt{3}p_{12}}{6} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{3} + \frac{\sqrt{3}p_9}{6}\end{bmatrix}$$

$$= \frac{\sqrt{3}\left(-p_{10} - 2p_{11} - p_{12} + p_7 + 2p_8 + p_9\right)}{6}$$

)

  Error in Orthogonalization of SALCs

- orbital of B2g with energy = $\tilde{\alpha}_F$
  (

$$\begin{bmatrix}0 & 1\end{bmatrix} * \begin{bmatrix}\text{1. SALC in B2g}\\ \text{2. SALC in B2g}\end{bmatrix} = \begin{bmatrix}0 & 1\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{6}p_1}{6} - \frac{\sqrt{6}p_2}{6} + \frac{\sqrt{6}p_3}{6} - \frac{\sqrt{6}p_4}{6} + \frac{\sqrt{6}p_5}{6} - \frac{\sqrt{6}p_6}{6}\\ -\frac{\sqrt{6}p_{10}}{6} + \frac{\sqrt{6}p_{11}}{6} - \frac{\sqrt{6}p_{12}}{6} + \frac{\sqrt{6}p_7}{6} - \frac{\sqrt{6}p_8}{6} + \frac{\sqrt{6}p_9}{6}\end{bmatrix}$$

$$= \frac{\sqrt{6}\left(-p_{10} + p_{11} - p_{12} + p_7 - p_8 + p_9\right)}{6}$$

)

- orbital of B2g with energy = $\tilde{\alpha} - 2\tilde{\beta}$
  (

$$\begin{bmatrix}1 & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in B2g}\\ \text{2. SALC in B2g}\end{bmatrix} = \begin{bmatrix}1 & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{6}p_1}{6} - \frac{\sqrt{6}p_2}{6} + \frac{\sqrt{6}p_3}{6} - \frac{\sqrt{6}p_4}{6} + \frac{\sqrt{6}p_5}{6} - \frac{\sqrt{6}p_6}{6}\\ -\frac{\sqrt{6}p_{10}}{6} + \frac{\sqrt{6}p_{11}}{6} - \frac{\sqrt{6}p_{12}}{6} + \frac{\sqrt{6}p_7}{6} - \frac{\sqrt{6}p_8}{6} + \frac{\sqrt{6}p_9}{6}\end{bmatrix}$$

$$= \frac{\sqrt{6}\left(p_1 - p_2 + p_3 - p_4 + p_5 - p_6\right)}{6}$$

)

- orbital of E2u with energy = 0
  (
  $$\begin{bmatrix}\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E2u}\\ \text{2. SALC in E2u}\\ \text{3. SALC in E2u}\\ \text{4. SALC in E2u}\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} -\\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} -\\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} +\\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} -\end{bmatrix}$$
  $$= 0$$
)
  Error in Orthogonalization of SALCsError in Orthogonalization of SALCsError in Orthogonalization of SALCs

- orbital of E2u with energy = $\tilde{\alpha}_F$
  (
  $$\begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E2u}\\ \text{2. SALC in E2u}\\ \text{3. SALC in E2u}\\ \text{4. SALC in E2u}\end{bmatrix} = \begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} - \frac{\sqrt{3}p_6}{6}\\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} - \frac{\sqrt{3}p_6}{6}\\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} + \frac{\sqrt{3}p_6}{3}\\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} - \frac{\sqrt{3}p_9}{3}\end{bmatrix}$$
  $$= \frac{\sqrt{3}\left(p_{10} + p_{11} - 2p_{12} + p_7 + p_8 - 2p_9\right)}{6}$$
)
  Error in Orthogonalization of SALCs

- orbital of E2u with energy = $\frac{3\tilde{\alpha}}{2} - \frac{3\tilde{\beta}}{2}$
  (
  $$\begin{bmatrix}-\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E2u}\\ \text{2. SALC in E2u}\\ \text{3. SALC in E2u}\\ \text{4. SALC in E2u}\end{bmatrix} = \begin{bmatrix}-\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} -\\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} -\\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} +\\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} -\end{bmatrix}$$
  $$= \frac{\sqrt{6}\left(-p_1 + p_2 - p_4 + p_5\right)}{4}$$
)
  Error in Orthogonalization of SALCs

- orbital of E2u with energy = $\frac{3\tilde{\alpha}}{2} - \frac{3\tilde{\beta}}{2}$
  (
  $$\begin{bmatrix}-\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E2u}\\ \text{2. SALC in E2u}\\ \text{3. SALC in E2u}\\ \text{4. SALC in E2u}\end{bmatrix} = \begin{bmatrix}-\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} -\\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} -\\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} +\\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} -\end{bmatrix}$$
  $$= \frac{\sqrt{6}\left(-p_1 + p_3 - p_4 + p_6\right)}{4}$$
)
  Error in Orthogonalization of SALCs

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 12E \oplus 4C'2 \oplus 12\sigma h \oplus 4\sigma v$$

$$\Gamma_{irred} = 2A1g \oplus 2E2g \oplus 2B1u \oplus 2E1u$$

- SALC of A1g: $\frac{\sqrt{6}(\mathrm{s}_1+\mathrm{s}_2+\mathrm{s}_3+\mathrm{s}_4+\mathrm{s}_5+\mathrm{s}_6)}{6}$
- SALC of A1g: $\frac{\sqrt{6}(\mathrm{s}_{10}+\mathrm{s}_{11}+\mathrm{s}_{12}+\mathrm{s}_7+\mathrm{s}_8+\mathrm{s}_9)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(2\mathrm{s}_1-\mathrm{s}_2-\mathrm{s}_3+2\mathrm{s}_4-\mathrm{s}_5-\mathrm{s}_6)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(-\mathrm{s}_1+2\mathrm{s}_2-\mathrm{s}_3-\mathrm{s}_4+2\mathrm{s}_5-\mathrm{s}_6)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(-\mathrm{s}_1-\mathrm{s}_2+2\mathrm{s}_3-\mathrm{s}_4-\mathrm{s}_5+2\mathrm{s}_6)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(\mathrm{s}_{10}+\mathrm{s}_{11}-2\mathrm{s}_{12}+\mathrm{s}_7+\mathrm{s}_8-2\mathrm{s}_9)}{6}$
- SALC of B1u: $\frac{\sqrt{6}(\mathrm{s}_1-\mathrm{s}_2+\mathrm{s}_3-\mathrm{s}_4+\mathrm{s}_5-\mathrm{s}_6)}{6}$
- SALC of B1u: $\frac{\sqrt{6}(-\mathrm{s}_{10}+\mathrm{s}_{11}-\mathrm{s}_{12}+\mathrm{s}_7-\mathrm{s}_8+\mathrm{s}_9)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(2\mathrm{s}_1+\mathrm{s}_2-\mathrm{s}_3-2\mathrm{s}_4-\mathrm{s}_5+\mathrm{s}_6)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(\mathrm{s}_1+2\mathrm{s}_2+\mathrm{s}_3-\mathrm{s}_4-2\mathrm{s}_5-\mathrm{s}_6)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(-\mathrm{s}_1+\mathrm{s}_2+2\mathrm{s}_3+\mathrm{s}_4-\mathrm{s}_5-2\mathrm{s}_6)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(-\mathrm{s}_{10}-2\mathrm{s}_{11}-\mathrm{s}_{12}+\mathrm{s}_7+2\mathrm{s}_8+\mathrm{s}_9)}{6}$

Put together these SALCs form the following Hamilton matrices:

$$H_{A1g} = \left[\begin{bmatrix} \tilde{\alpha}_s + 2\tilde{\beta}_s & 6beta_{\mathrm{s}_F} \\ 6beta_{\mathrm{s}_F} & alpha_{\mathrm{s}_F} \end{bmatrix}\right]$$

$$H_{E2g} = \left[\begin{bmatrix} \tilde{\alpha}_s - \tilde{\beta}_s & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & 0 \\ -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s - \tilde{\beta}_s & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & 0 \\ -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s - \tilde{\beta}_s & 0 \\ 0 & 0 & 0 & alpha_{\mathrm{s}_F} \end{bmatrix}\right]$$

$$H_{B1u} = \left[\begin{bmatrix} \tilde{\alpha}_s - 2\tilde{\beta}_s & 0 \\ 0 & alpha_{\mathrm{s}_F} \end{bmatrix}\right]$$

$$H_{E1u} = \left[\begin{bmatrix} \tilde{\alpha}_s + \tilde{\beta}_s & \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & -\frac{\tilde{\alpha}_s}{2} - \frac{\tilde{\beta}_s}{2} & 0 \\ \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s + \tilde{\beta}_s & \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & 0 \\ -\frac{\tilde{\alpha}_s}{2} - \frac{\tilde{\beta}_s}{2} & \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s + \tilde{\beta}_s & 0 \\ 0 & 0 & 0 & alpha_{\mathrm{s}_F} \end{bmatrix}\right]$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A1g with energy = $\frac{\tilde{\alpha}_s}{2}+\frac{alpha_{\mathsf{s}_F}}{2}+\tilde{\beta}_s-\frac{\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}_s+alpha^2_{\mathsf{s}_F}-4alpha_{\mathsf{s}_F}\tilde{\beta}_s+4\tilde{\beta}_s^2+144beta^2_{\mathsf{s}_F}}}{2}$

  (

  $$\left[-\frac{12beta_{\mathsf{s}_F}}{\sqrt{144\left|\frac{beta_{\mathsf{s}_F}}{\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}_s+alpha^2_{\mathsf{s}_F}-4alpha_{\mathsf{s}_F}\tilde{\beta}_s+4\tilde{\beta}_s^2+144beta^2_{\mathsf{s}_F}}}\right|^2+1\left(\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+\cdots\right.}}\right.$$

  $$=\frac{\sqrt{6}\left(-12beta_{\mathsf{s}_F}\left(\mathsf{s}_1+\mathsf{s}_2+\mathsf{s}_3+\mathsf{s}_4+\mathsf{s}_5+\mathsf{s}_6\right)+\left(\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}\cdots\right.\right.}{6\sqrt{144\left|\frac{beta_{\mathsf{s}_F}}{\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s+\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}_s+alpha^2_{\mathsf{s}_F}-4alpha_{\mathsf{s}_F}\tilde{\beta}_s+4\tilde{\beta}_s^2+144beta^2_{\mathsf{s}_F}}}\right|^2+1\left(\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\cdots\right.}}$$

)

- orbital of A1g with energy = $\frac{\tilde{\alpha}_s}{2}+\frac{alpha_{\mathsf{s}_F}}{2}+\tilde{\beta}_s+\frac{\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}_s+alpha^2_{\mathsf{s}_F}-4alpha_{\mathsf{s}_F}\tilde{\beta}_s+4\tilde{\beta}_s^2+144beta^2_{\mathsf{s}_F}}}{2}$

  (

  $$\left[-\frac{12beta_{\mathsf{s}_F}}{\sqrt{144\left|\frac{beta_{\mathsf{s}_F}}{\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s-\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}_s+alpha^2_{\mathsf{s}_F}-4alpha_{\mathsf{s}_F}\tilde{\beta}_s+4\tilde{\beta}_s^2+144beta^2_{\mathsf{s}_F}}}\right|^2+1\left(\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s-\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+\cdots\right.}}\right.$$

  $$=\frac{\sqrt{6}\left(-12beta_{\mathsf{s}_F}\left(\mathsf{s}_1+\mathsf{s}_2+\mathsf{s}_3+\mathsf{s}_4+\mathsf{s}_5+\mathsf{s}_6\right)+\left(\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s-\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}\cdots\right.\right.}{6\sqrt{144\left|\frac{beta_{\mathsf{s}_F}}{\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\tilde{\beta}_s-\sqrt{\tilde{\alpha}_s^2-2\tilde{\alpha}_s alpha_{\mathsf{s}_F}+4\tilde{\alpha}_s\tilde{\beta}_s+alpha^2_{\mathsf{s}_F}-4alpha_{\mathsf{s}_F}\tilde{\beta}_s+4\tilde{\beta}_s^2+144beta^2_{\mathsf{s}_F}}}\right|^2+1\left(\tilde{\alpha}_s-alpha_{\mathsf{s}_F}+2\cdots\right.}}$$

)

- orbital of E1u with energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$

  (

  $$\begin{bmatrix}\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0\end{bmatrix}*\begin{bmatrix}\text{1. SALC in E1u}\\ \text{2. SALC in E1u}\\ \text{3. SALC in E1u}\\ \text{4. SALC in E1u}\end{bmatrix}=\begin{bmatrix}\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0\end{bmatrix}*\left[\begin{matrix}\frac{\sqrt{3}\mathsf{s}_1}{3}+\frac{\sqrt{3}\mathsf{s}_2}{6}-\frac{\sqrt{3}\mathsf{s}_3}{6}-\frac{\sqrt{3}\mathsf{s}_4}{3}-\frac{\sqrt{3}\mathsf{s}_5}{6}+\cdots\\ \frac{\sqrt{3}\mathsf{s}_1}{6}+\frac{\sqrt{3}\mathsf{s}_2}{3}+\frac{\sqrt{3}\mathsf{s}_3}{6}-\frac{\sqrt{3}\mathsf{s}_4}{6}-\frac{\sqrt{3}\mathsf{s}_5}{3}-\cdots\\ -\frac{\sqrt{3}\mathsf{s}_1}{6}+\frac{\sqrt{3}\mathsf{s}_2}{6}+\frac{\sqrt{3}\mathsf{s}_3}{3}+\frac{\sqrt{3}\mathsf{s}_4}{6}-\frac{\sqrt{3}\mathsf{s}_5}{6}-\cdots\\ -\frac{\sqrt{3}\mathsf{s}_{10}}{6}-\frac{\sqrt{3}\mathsf{s}_{11}}{3}-\frac{\sqrt{3}\mathsf{s}_{12}}{6}+\frac{\sqrt{3}\mathsf{s}_7}{6}+\frac{\sqrt{3}\mathsf{s}_8}{3}+\cdots\end{matrix}\right.$$

  $$=\frac{\sqrt{6}\left(\mathsf{s}_1+\mathsf{s}_2-\mathsf{s}_4-\mathsf{s}_5\right)}{4}$$

)

  Error in Orthogonalization of SALCs

- orbital of E1u with energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$

  (

  $$\begin{bmatrix}-\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0\end{bmatrix}*\begin{bmatrix}\text{1. SALC in E1u}\\ \text{2. SALC in E1u}\\ \text{3. SALC in E1u}\\ \text{4. SALC in E1u}\end{bmatrix}=\begin{bmatrix}-\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0\end{bmatrix}*\left[\begin{matrix}\frac{\sqrt{3}\mathsf{s}_1}{3}+\frac{\sqrt{3}\mathsf{s}_2}{6}-\frac{\sqrt{3}\mathsf{s}_3}{6}-\frac{\sqrt{3}\mathsf{s}_4}{3}-\frac{\sqrt{3}\mathsf{s}_5}{6}+\cdots\\ \frac{\sqrt{3}\mathsf{s}_1}{6}+\frac{\sqrt{3}\mathsf{s}_2}{3}+\frac{\sqrt{3}\mathsf{s}_3}{6}-\frac{\sqrt{3}\mathsf{s}_4}{6}-\frac{\sqrt{3}\mathsf{s}_5}{3}-\cdots\\ -\frac{\sqrt{3}\mathsf{s}_1}{6}+\frac{\sqrt{3}\mathsf{s}_2}{6}+\frac{\sqrt{3}\mathsf{s}_3}{3}+\frac{\sqrt{3}\mathsf{s}_4}{6}-\frac{\sqrt{3}\mathsf{s}_5}{6}-\cdots\\ -\frac{\sqrt{3}\mathsf{s}_{10}}{6}-\frac{\sqrt{3}\mathsf{s}_{11}}{3}-\frac{\sqrt{3}\mathsf{s}_{12}}{6}+\frac{\sqrt{3}\mathsf{s}_7}{6}+\frac{\sqrt{3}\mathsf{s}_8}{3}\cdots\end{matrix}\right.$$

  $$=\frac{\sqrt{6}\left(-\mathsf{s}_1+\mathsf{s}_3+\mathsf{s}_4-\mathsf{s}_6\right)}{4}$$

)

  Error in Orthogonalization of SALCs

- orbital of E1u with energy = 0
  (
  $$\begin{bmatrix}\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E1u}\\ \text{2. SALC in E1u}\\ \text{3. SALC in E1u}\\ \text{4. SALC in E1u}\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}s_1}{3} + \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{3} - \\ \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} + \frac{\sqrt{3}s_4}{6} - \\ -\frac{\sqrt{3}s_{10}}{6} - \frac{\sqrt{3}s_{11}}{3} - \frac{\sqrt{3}s_{12}}{6} + \frac{\sqrt{3}s_7}{6} + \end{bmatrix}$$
  $$= 0$$
)
  Error in Orthogonalization of SALCsError in Orthogonalization of SALCsError in Orthogonalization of SALCs

- orbital of E1u with energy = $alpha_{s_F}$
  (
  $$\begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E1u}\\ \text{2. SALC in E1u}\\ \text{3. SALC in E1u}\\ \text{4. SALC in E1u}\end{bmatrix} = \begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}s_1}{3} + \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{6}\\ \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{3} - \frac{\sqrt{3}s_6}{6}\\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} + \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} - \frac{\sqrt{3}s_6}{3}\\ -\frac{\sqrt{3}s_{10}}{6} - \frac{\sqrt{3}s_{11}}{3} - \frac{\sqrt{3}s_{12}}{6} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{3} + \frac{\sqrt{3}s_9}{6}\end{bmatrix}$$
  $$= \frac{\sqrt{3}\left(-s_{10} - 2s_{11} - s_{12} + s_7 + 2s_8 + s_9\right)}{6}$$
)
  Error in Orthogonalization of SALCs

- orbital of E2g with energy = 0
  (
  $$\begin{bmatrix}\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E2g}\\ \text{2. SALC in E2g}\\ \text{3. SALC in E2g}\\ \text{4. SALC in E2g}\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} - \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} - \\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} + \\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} - \end{bmatrix}$$
  $$= 0$$
)
  Error in Orthogonalization of SALCsError in Orthogonalization of SALCsError in Orthogonalization of SALCs

- orbital of E2g with energy = $alpha_{s_F}$
  (
  $$\begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\text{1. SALC in E2g}\\ \text{2. SALC in E2g}\\ \text{3. SALC in E2g}\\ \text{4. SALC in E2g}\end{bmatrix} = \begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} - \frac{\sqrt{3}s_6}{6}\\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} - \frac{\sqrt{3}s_6}{6}\\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{3}\\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} - \frac{\sqrt{3}s_9}{3}\end{bmatrix}$$
  $$= \frac{\sqrt{3}\left(s_{10} + s_{11} - 2s_{12} + s_7 + s_8 - 2s_9\right)}{6}$$
)
  Error in Orthogonalization of SALCs

- orbital of E2g with energy = $\frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}_s}{2}$
  (

$$\left[-\frac{\sqrt{2}}{2} \quad \frac{\sqrt{2}}{2} \quad 0 \quad 0\right] * \begin{bmatrix} \text{1. SALC in E2g} \\ \text{2. SALC in E2g} \\ \text{3. SALC in E2g} \\ \text{4. SALC in E2g} \end{bmatrix} = \left[-\frac{\sqrt{2}}{2} \quad \frac{\sqrt{2}}{2} \quad 0 \quad 0\right] * \begin{bmatrix} \frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} - \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} - \\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} + \\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} - \end{bmatrix}$$

$$= \frac{\sqrt{6}\left(-s_1 + s_2 - s_4 + s_5\right)}{4}$$

)
  Error in Orthogonalization of SALCs

- orbital of E2g with energy = $\frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}_s}{2}$
  (

$$\left[-\frac{\sqrt{2}}{2} \quad 0 \quad \frac{\sqrt{2}}{2} \quad 0\right] * \begin{bmatrix} \text{1. SALC in E2g} \\ \text{2. SALC in E2g} \\ \text{3. SALC in E2g} \\ \text{4. SALC in E2g} \end{bmatrix} = \left[-\frac{\sqrt{2}}{2} \quad 0 \quad \frac{\sqrt{2}}{2} \quad 0\right] * \begin{bmatrix} \frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} - \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} - \\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} + \\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} - \end{bmatrix}$$

$$= \frac{\sqrt{6}\left(-s_1 + s_3 - s_4 + s_6\right)}{4}$$

)
  Error in Orthogonalization of SALCs

- orbital of B1u with energy = $alpha_{s_F}$
  (

$$\left[0 \quad 1\right] * \begin{bmatrix} \text{1. SALC in B1u} \\ \text{2. SALC in B1u} \end{bmatrix} = \left[0 \quad 1\right] * \begin{bmatrix} \frac{\sqrt{6}s_1}{6} - \frac{\sqrt{6}s_2}{6} + \frac{\sqrt{6}s_3}{6} - \frac{\sqrt{6}s_4}{6} + \frac{\sqrt{6}s_5}{6} - \frac{\sqrt{6}s_6}{6} \\ -\frac{\sqrt{6}s_{10}}{6} + \frac{\sqrt{6}s_{11}}{6} - \frac{\sqrt{6}s_{12}}{6} + \frac{\sqrt{6}s_7}{6} - \frac{\sqrt{6}s_8}{6} + \frac{\sqrt{6}s_9}{6} \end{bmatrix}$$

$$= \frac{\sqrt{6}\left(-s_{10} + s_{11} - s_{12} + s_7 - s_8 + s_9\right)}{6}$$

)

- orbital of B1u with energy = $\tilde{\alpha}_s - 2\tilde{\beta}_s$
  (

$$\left[1 \quad 0\right] * \begin{bmatrix} \text{1. SALC in B1u} \\ \text{2. SALC in B1u} \end{bmatrix} = \left[1 \quad 0\right] * \begin{bmatrix} \frac{\sqrt{6}s_1}{6} - \frac{\sqrt{6}s_2}{6} + \frac{\sqrt{6}s_3}{6} - \frac{\sqrt{6}s_4}{6} + \frac{\sqrt{6}s_5}{6} - \frac{\sqrt{6}s_6}{6} \\ -\frac{\sqrt{6}s_{10}}{6} + \frac{\sqrt{6}s_{11}}{6} - \frac{\sqrt{6}s_{12}}{6} + \frac{\sqrt{6}s_7}{6} - \frac{\sqrt{6}s_8}{6} + \frac{\sqrt{6}s_9}{6} \end{bmatrix}$$

$$= \frac{\sqrt{6}\left(s_1 - s_2 + s_3 - s_4 + s_5 - s_6\right)}{6}$$

)

Transitions from p orbitals into s orbitals

State with Occupation [2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2] (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2}:

- $-[2,2,2,0,0,0,2,2,2,2,2,2],(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,1,2,0,0,0,2,2,2,2,2,2),(0,1,0,0,0,0$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $16\tilde{\alpha}+4\tilde{\alpha}_F$
- Energy of State after Excitation: $\frac{29\tilde{\alpha}}{2}+4\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3}:

- $-[2,2,2,0,0,0,2,2,2,2,2,2],(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,2,1,0,0,0,2,2,2,2,2,2),(0,0,1,0,0,0$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $16\tilde{\alpha}+4\tilde{\alpha}_F$
- Energy of State after Excitation: $\frac{29\tilde{\alpha}}{2}+4\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2) (4 unpaired electrons, mult 5.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,1,1,1,1,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,0,1,1,1,0,2,2,2,2,2,2),(0,1,0,0,0,$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$

- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $13\tilde{\alpha}+5\tilde{\alpha}_F-3\tilde{\beta}$
- Energy of State after Excitation: $\frac{23\tilde{\alpha}}{2}+5\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-\frac{9\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into $\psi_{\mathbf{s}_3}$:

- $-(2,1,1,1,1,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0)\rightarrow(2,1,0,1,1,0,2,2,2,2,2,2),(0,0,1,0,0,$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2\mathrm{p}_1-\mathrm{p}_2+\mathrm{p}_3+2\mathrm{p}_4+\mathrm{p}_5-\mathrm{p}_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2\mathrm{s}_1-\mathrm{s}_2+\mathrm{s}_3+2\mathrm{s}_4+\mathrm{s}_5-\mathrm{s}_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $13\tilde{\alpha}+5\tilde{\alpha}_F-3\tilde{\beta}$
- Energy of State after Excitation: $\frac{23\tilde{\alpha}}{2}+5\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-\frac{9\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into $\psi_{\mathbf{s}_2}$:

- $-(2,1,2,0,1,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0)\rightarrow(2,0,2,0,1,0,2,2,2,2,2,2),(0,1,0,0,0,$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2\mathrm{p}_1+\mathrm{p}_2-\mathrm{p}_3-2\mathrm{p}_4-\mathrm{p}_5+\mathrm{p}_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2\mathrm{s}_1+\mathrm{s}_2-\mathrm{s}_3-2\mathrm{s}_4-\mathrm{s}_5+\mathrm{s}_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2}+4\tilde{\alpha}_F-\frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha}+4\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-3\tilde{\beta}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into $\psi_{\mathbf{s}_3}$:

- $-(2,1,2,0,1,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0)\rightarrow(2,1,1,0,1,0,2,2,2,2,2,2),(0,0,1,0,0,$

- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2}+4\tilde{\alpha}_F-\frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha}+4\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-3\tilde{\beta}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{sF}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,1,2,1,0,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,0,2,1,0,0,2,2,2,2,2,2),(0,1,0,0,0,$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2}+5\tilde{\alpha}_F-\frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha}+5\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-3\tilde{\beta}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{sF}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3}:

- $-(2,1,2,1,0,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,1,1,1,0,0,2,2,2,2,2,2),(0,0,1,0,0,$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2}+5\tilde{\alpha}_F-\frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha}+5\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-3\tilde{\beta}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{sF}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,2,1,0,1,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,1,1,0,1,0,2,2,2,2,2,2),(0,1,0,0,0,$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2\mathsf{p}_1+\mathsf{p}_2-\mathsf{p}_3-2\mathsf{p}_4-\mathsf{p}_5+\mathsf{p}_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2\mathsf{s}_1+\mathsf{s}_2-\mathsf{s}_3-2\mathsf{s}_4-\mathsf{s}_5+\mathsf{s}_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2}+4\tilde{\alpha}_F-\frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha}+4\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-3\tilde{\beta}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3}:

- $-(2,2,1,0,1,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,2,0,0,1,0,2,2,2,2,2,2),(0,0,1,0,0,$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2\mathsf{p}_1-\mathsf{p}_2+\mathsf{p}_3+2\mathsf{p}_4+\mathsf{p}_5-\mathsf{p}_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2\mathsf{s}_1-\mathsf{s}_2+\mathsf{s}_3+2\mathsf{s}_4+\mathsf{s}_5-\mathsf{s}_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2}+4\tilde{\alpha}_F-\frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha}+4\tilde{\alpha}_F+\frac{3\tilde{\alpha}_s}{2}-3\tilde{\beta}+\frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_F}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,2,1,1,0,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,1,1,1,0,0,2,2,2,2,2,2),(0,1,0,0,0,$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2\mathsf{p}_1+\mathsf{p}_2-\mathsf{p}_3-2\mathsf{p}_4-\mathsf{p}_5+\mathsf{p}_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2\mathsf{s}_1+\mathsf{s}_2-\mathsf{s}_3-2\mathsf{s}_4-\mathsf{s}_5+\mathsf{s}_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2}+\frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2}+\frac{3\tilde{\alpha}_s}{2}-\frac{3\tilde{\beta}}{2}+\frac{3\tilde{\beta}_s}{2}$

- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 5\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{alpha_{\mathsf{s}_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into $\psi_{\mathbf{s}_3}$:

- $-(2,2,1,1,0,0,2,2,2,2,2,2),(0,0,0,0,0,0,0,0,0,0,0,0) \rightarrow (2,2,0,1,0,0,2,2,2,2,2,2),(0,0,1,0,0,$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2\mathsf{p}_1 - \mathsf{p}_2 + \mathsf{p}_3 + 2\mathsf{p}_4 + \mathsf{p}_5 - \mathsf{p}_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2\mathsf{s}_1 - \mathsf{s}_2 + \mathsf{s}_3 + 2\mathsf{s}_4 + \mathsf{s}_5 - \mathsf{s}_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 5\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{alpha_{\mathsf{s}_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

\- AND

Dispersion energies following from the transitions

1. State: A = [2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2] ↔ B = (2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{54\tilde{\delta}^4}{r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

2. State: A = (2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = [2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2]

$$E_{dispersion} = -\frac{54\tilde{\delta}^4}{r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

3. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

4. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

5. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

6. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

7. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

8. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

9. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

10. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

11. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

12. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

13. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

14. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

15. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

16. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

17. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

18. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$

Ground-State:
A =[2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2]↔ B =[2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2]

$$E_{dispersion} = -\frac{27\tilde{\delta}^4}{r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$